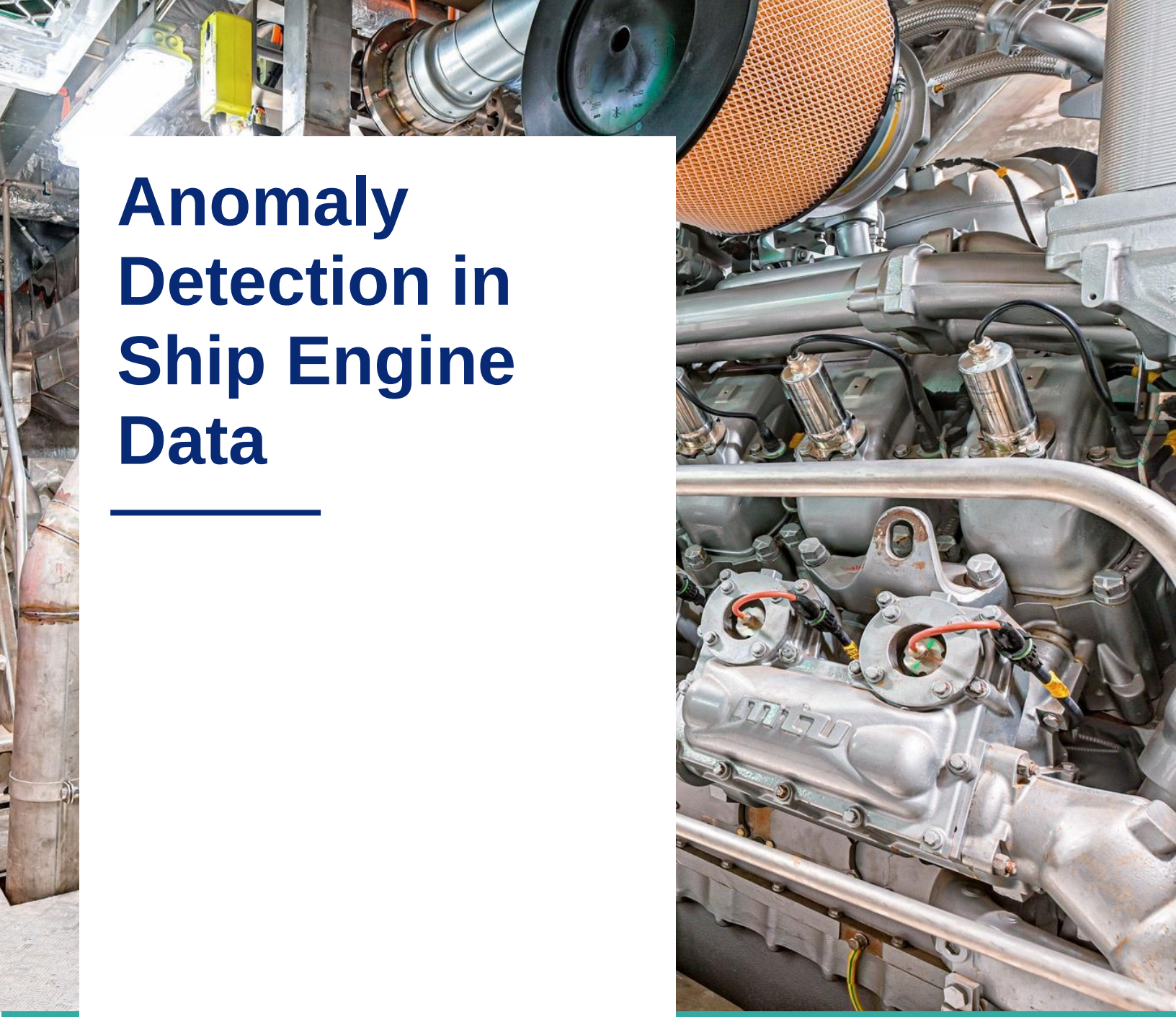
Anomaly Detection in Ship Engine Data



Flex LNG Ltd. (2021)

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C101 – Week 5 Mini-Project

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Context and Problem statement

Modern ship engines collect data on variables including RPMs, pressure and temperature of different sections, among others.

Issues with engines can lead to malfunctions, safety hazards, and downtime, resulting in the breakdown of a ship's overall functionality, consequently impacting your business, such as affecting revenue via failure to deliver goods. To address this risk, unsupervised anomaly detection techniques can be used to predict these issues.

By predicting timely maintenance, the business will be able to increase profit by reducing downtime, limiting fuel consumption, and increasing customer satisfaction through timely deliveries.

Methodology

Before looking towards anomaly detection, I went to understand the data. The data provided consisted of 19,535 observations, each with 6 features. The data had no missing or duplicate data, and consisted of solely numerical data. After reviewing the descriptive statistics, I visualized the variables using histograms and box plots to gain a deeper understanding of their distributions and possible outliers.

To create a viable anomaly detection system, we used three separate methods: (1) statistical anomaly detection through interquartile range, (2) One Class Support Vector Machines and (3) Isolation Forest. Through adjusting of certain parameters, we could achieve the desired proportion of anomalies within the data of 1-5%.

Table 1. Anomaly detection method summary

Anomaly detection system	Description	Adjusted Parameters
Interquartile Range (IQR)	Statistical method that flags observations outside bounds defined as below Q1 – 1.5*IQR or above Q3 + 1.5*IQR	# of variables that fell outside of defined bounds within one observation
One Class SVM (OCSVM)	Learns the boundary of normal engine behavior in multivariate space and flags points outside this boundary as anomalies.	*nu* (expected anomaly proportion) *gamma* (boundary flexibility)
Isolation Forest (IF)	Tree-based method that isolates anomalies through random feature splits, leading to anomalies being isolated more quickly than normal points.	Contamination rate (% of data to be classified as anomalies)

For the IQR methodology, I used the standard statistical bounds to determine anomalies for individual features however, for this anomaly detection task, we primarily care about anomalies as whole observations as opposed to only within a feature. I therefore adjusted the number of outliers within an observation needed to hit this 1-5% range.

Since both OCSVM and IF are machine learning anomaly detection methods, this involved a less manual approach where I simply adjusted the parameters in *Table 1* and then applied a principal component analysis

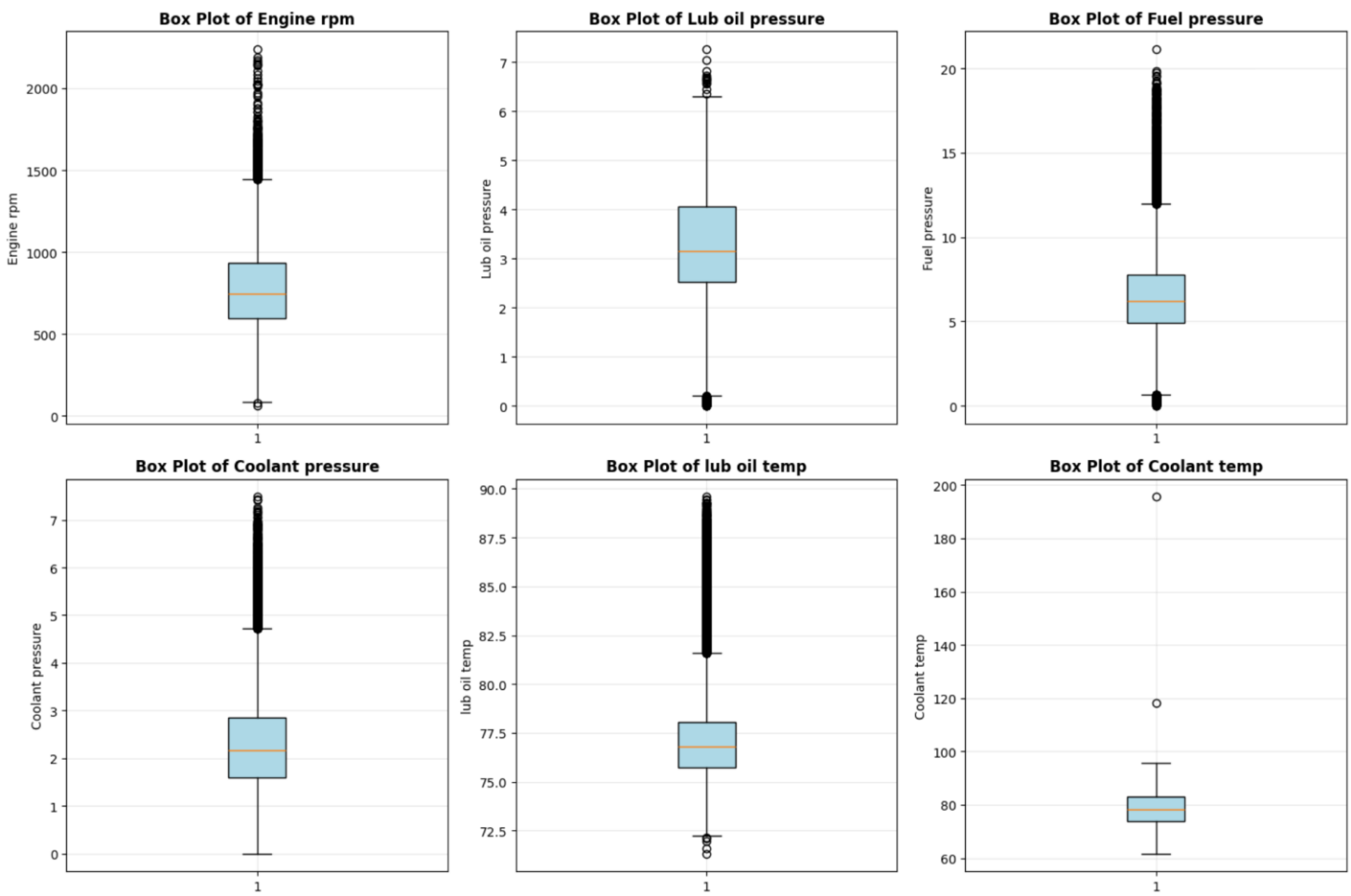
(PCA) for each in order to visualize the outputs of each of method. Even though PCA uses the same 2 components in the 2D visualization, the differences in method result can be seen through the highlighting of difference inliers and outliers in their respective graphs.

Results

IQR

While the boxplots were created to help me understand initial patterns in the data, these diagrams were a visual representation of the first step of the IQR method I would implement.

Figure 1. Box plot of each feature



The box plot diagrams showed and initial estimation of outliers in each feature, with engine RPM (2.38%), fuel pressure (5.81%), coolant pressure (4.02%), and lubricant oil temperature (13.40%) as the variables with the most outliers. After finding the number and percent of observations that had one, two and three features marked as outliers, “at least 2” outliers per observation led to a sufficing result of classifying 2.16% (422 observations) of the data as anomalies. If our within-observation-outlier-number is “at least one”, our anomaly percentage would be too high at 23.73%; if it was “at least 3”, less than 0.01%. Neither of these were satisfactory for the 1-5% band.

Table 2. IQR Parameter adjustments

Minimum number of outlier features per observation	Number of observations flagged	Percentage of dataset	Assessment vs 1–5% target
At least 1 outlier	4’696	23.73%	Too high (over-flags anomalies)
At least 2 outliers	422	2.16%	Appropriate (within target range)
At least 3 outliers	11	<0.01%	Too low (under-detects anomalies)

OCSVM

With OCSVM, after trial of three separate variable states, a nu of 0.03 and gamma of 0.5 led to a outlier classification percentage of 3.77%. This state was later used in the PCA visualisation.

Table 3. OCSVM Parameter adjustments

Trial (nu, gamma)	Number of outliers	Percentage of dataset	Assessment vs 1–5% target
Trial 1 (nu = 0.05, γ = 0.5)	1,043	5.34%	Slightly above target
Trial 2 (nu = 0.03, γ = 0.5)	737	3.77%	Within target range
Trial 3 (nu = 0.05, γ = 0.1)	978	5.01%	Borderline high

IF

With IF, we can explicitly pre-define % of outliers through contamination values, so anything between 1-5% would have been acceptable for this system, however, for the purpose of comparing PCA visualisations between the two methods, a contamination value of 3.77% was used.

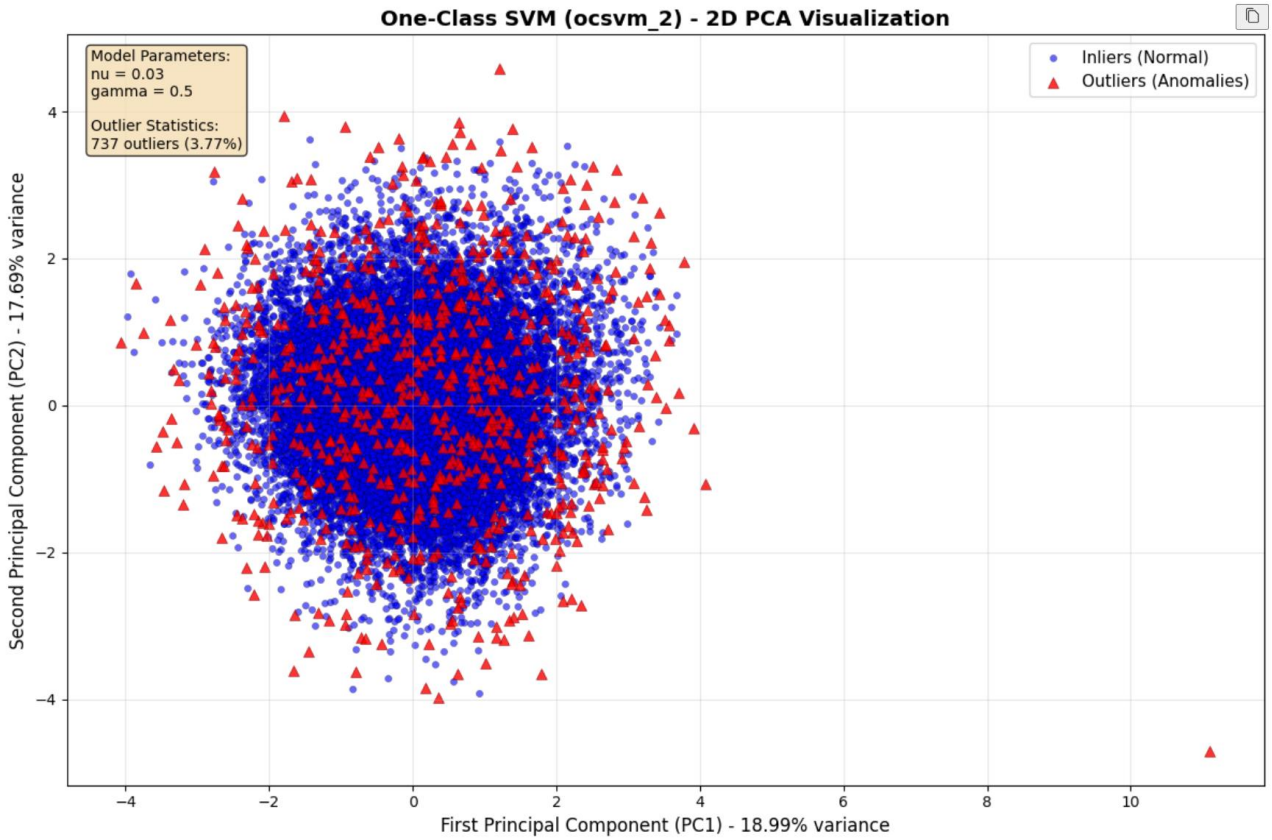
PCA

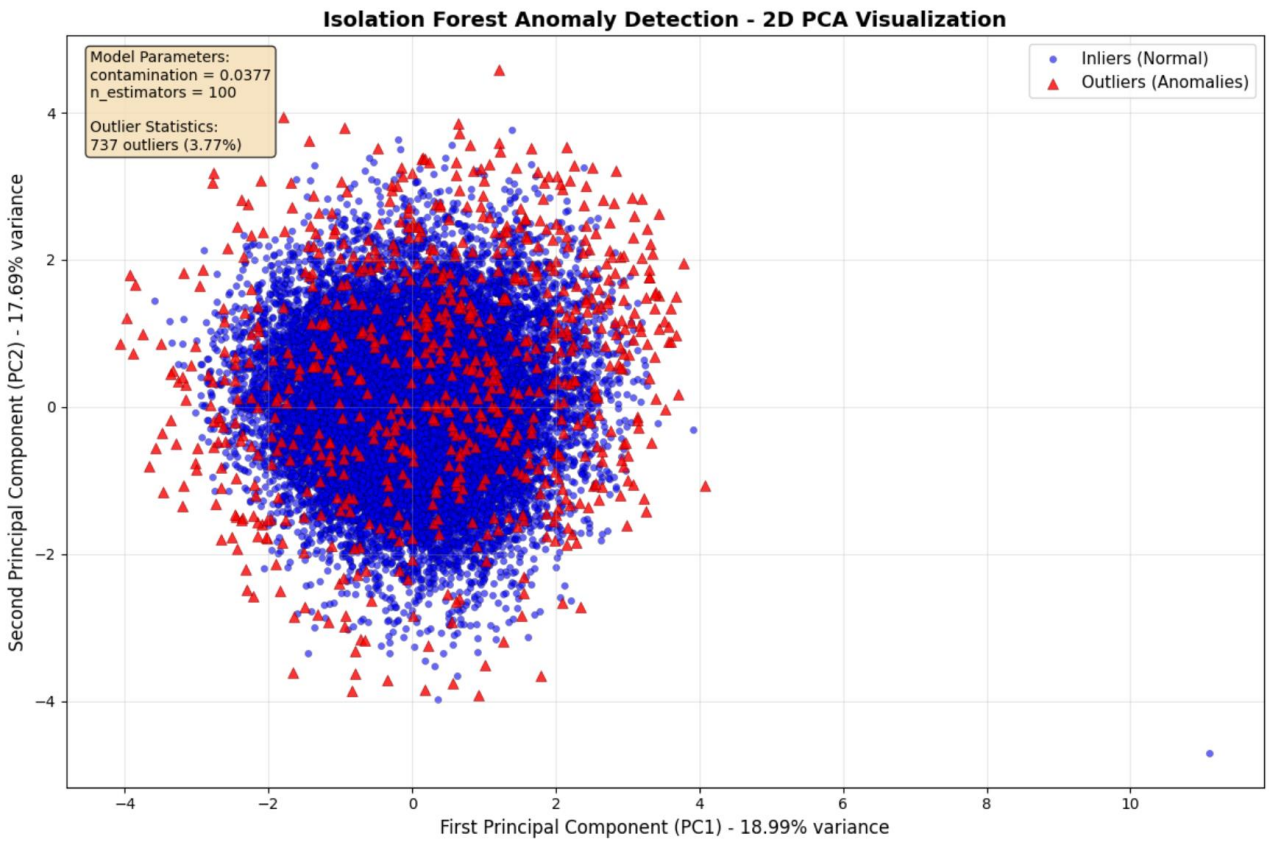
After putting both OCSVM and IF into separate PCA visualizations we could see a soft trend that outliers tended to be closer to the edge of the clusters; however, there were many outliers in the middle and many inliers towards the edge in both graphs. Additionally explained variance of each component was low at a 19% and 18%.

IF PCA has less density of outliers in the middle of the graph vs OCSVM. Though, this "thumb in the air estimate" is not a particularly reliable method for comparing similar graphs such as these.

PCA is often difficult to translate into actionable insights because each axis represents a linear combination of multiple original features, rather than any single feature from the initial dataset. Therefore, deeper or alternative analysis would be required to extract more learnings from this step.

Figure 2. PCA graphs of OCSVM & IF





Conclusions

From the three methods of anomaly detection applied, all were able to match our required 1-5% benchmark of proportion of anomalies within the data, indicating all three as feasible methodologies. However, ML models are strongly preferred as using the IQR method identified extreme values in individual features, not multivariate anomalies effectively.

When comparing the ML models, IF led to slightly better outcomes when comparing PCA. Additionally, the contamination parameter makes outlier percentage very easy to adjust. Due to both these factors, IF is the recommended system in predicting outliers.

This model should now be analyzed by industry experts to determine whether outlier occurrences match engine issue occurrences well, and if so, incorporate this into their maintenance scheduling. If implemented, the business will be able to increase profit by reducing downtime, reducing safety risks for the crew, limiting fuel consumption, and increasing customer satisfaction through timely deliveries.

References

Devabrat, M., 2022. Predictive Maintenance on Ship's Main Engine using AI. Available at: https://dx.doi.org/10.21227/g3za-v415. [Accessed 3 January 2026]

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